



Mitigation of harmful algal blooms using iron-modified clays inside the semi-permeable membrane tubes

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ARTICLE INFO

Keywords:

Harmful algal blooms
Phosphate
Clay
Modification
MC-LR
Semi-permeable membrane

ABSTRACT

This study explored innovative methods for mitigating harmful algal blooms (HABs) using iron and chitosan-modified clay to deposit cyanobacteria cells, immobilize phosphate, and adsorb algal toxin microcystin LR (MC-LR). Compared with modified kaolinite, modified sepiolite showed superior capacity in phosphate removal at the same initial PO_4^{3-} concentrations. The maximum capacities of modified kaolinite and sepiolite for PO_4^{3-} adsorption were 20.7 and 30.7 mg PO_4^{3-} /g clay. The MC-LR removal by modified sepiolite was 95.5 % - 99.9 % when initial MC-LR concentrations ranged from 20 to 200 $\mu\text{g/L}$. Modified kaolinite could remove MC-LR by 70 % only when the MC-LR concentration was high (100–400 $\mu\text{g/L}$). Precipitated phosphate in clay could eventually be released back into the water body. Thus, modified clays packed in semi-permeable membrane tubes were used to recover phosphate from water bodies to minimize the internal cycling of phosphate. The PO_4^{3-} and MC-LR removal by the clays packed in the semi-permeable membrane tubes was much slower, but over 90 % removal of phosphate and microcystin could be achieved in 10 days. Considering its low cost, excellent performance, and eco-friendly properties, the proposed solution has the potential to be used for HAB control and prevention on a large scale.

1. Introduction

Algal blooms are one of the primary reasons for water quality deterioration and malodor development, which damage the aesthetic beauty of recreational waters and pose potential carcinogenic risks due to algal toxins [1–3]. Cyanobacteria harmful algal blooms (cHABs) are the most notorious toxin-releasing blooms in water bodies [4,5]. Cyanobacteria are prokaryotic organisms that are also known as blue-green algae. Toxins produced by cyanobacteria are known as cyanotoxins. Cyanotoxins can cause a wide range of human health effects, from skin irritation and gastrointestinal illness to chronic disease and even lethal poisoning [2,4,6,7]. Microcystins are the most widespread cyanotoxins. Microcystin-LR (MC-LR) is considered the most toxic cyanotoxin out of at least 80 identified microcystins [8].

Nutrient enrichment due to anthropogenic activities, climatic shifts, or the transportation of algal species via ship ballast water are a few probable causes of the massive expansion of HABs [1,9]. Evidence shows that HABs are often stimulated by combined phosphorus (P) and nitrogen (N) enrichment. Historically, P is considered the most significant nutrient in shaping the dynamics of freshwater algal productivity,

including HABs. In the United States and Europe, water quality policy related to eutrophication has been directed essentially toward P control in lakes and estuaries [10,11]. However, lakes and reservoirs often show little to no signs of recovery over the years following external P load reduction due to the internal cycling of P. This is related to the fact that P released from sediments becomes biologically available (e.g., dissolved reactive phosphorus – DRP), thus making it more accessible to primary producers than external P loads. When controlling HABs via external phosphorus control, internal P loading due to release from sediments and lysed algae should also be addressed [12,13].

Given the detrimental effects of HABs, it is essential to develop and implement effective remedies for this ecological challenge. Developing algal bloom mitigation strategies that are highly efficient, environmentally sustainable, economically viable, and user-friendly is essential. HAB managing methods are either preventive methods or mitigating methods [9]. Furthermore, these HAB controlling methods can be grouped as physical, biological, and chemical approaches [14,15]. Physical approaches, such as the use of natural clays or ultrasound, can disrupt algal cells or cause their sedimentation; however, they are often limited by low efficiency, an inability to remove phosphate, or high

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<https://doi.org/10.1016/j.algal.2025.104302>

Received 30 April 2025; Received in revised form 28 August 2025; Accepted 12 September 2025

Available online 13 September 2025

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energy demands [14,16,17]. Biological methods of HAB control utilize organisms such as viruses, bacteria, parasites, zooplankton, and crustaceans that can kill, dissolve, or remove algal cells [18,19]. The difficulties in the mass culture of these agents and the risks of introducing invasive species make it a less favorable option [20]. Chemical methods for algal bloom treatment include chemical coagulation, flocculation, algicides, hydrogen peroxide, hydroxide radicals, ozone, copper sulfate, and disinfectants [21–23]. Even though these inexpensive chemical approaches quickly respond to the problem, they pose health hazards and other substantial undesired impacts on co-occurring organisms [23]. Comprehensive studies on the effectiveness of every chemical approach, toxicity impacts, and general safety risks are needed to evaluate the benefits of the chemical methods [24].

Among all the methods for HAB control, clay-based methods are favored due to their low environmental impacts, higher removal efficiency, quick response, and low cost [17,25,26]. Clay acts as a ballast to the organisms, promoting cell sinking regardless of the organisms' motility and buoyancy [17,25]. Over the past few decades, clays have been identified and studied for their unique features in HAB treatment. Pan et al. have comprehensively studied the kinetic and equilibrium properties of the flocculation of 26 different natural clays and minerals in removing *M. aeruginosa* cells [27]. Zou et al. further continued this study, modifying the selected clays and minerals with chitosan, polyacrylamide (PAM), or $\text{NH}_4\text{Fe}(\text{SO}_4)_2$ separately, and evaluated their flocculation and removal efficiency of *M. aeruginosa* cells after modification [28]. Lanthanum-modified bentonite clay (Phoslock®) was developed to remove excess phosphorus from water bodies, although its high cost limited its application [29]. However, existing studies have focused only on removing *M. aeruginosa* cells or P removal, without investigating their performance in cyanobacteria, P, and cyanotoxin removal as a whole. In our study, modified clays with both chitosan and $\text{NH}_4\text{Fe}(\text{SO}_4)_2$ modification were developed to remove *M. aeruginosa* cells, phosphate, and the cyanotoxin MCLR, providing a wholesome HAB mitigation solution. HAB control efficiency was expected to be enhanced by modified clays through precipitating algal cells and immobilizing P and MC-LR from water. Chitosan is a natural, non-toxic, biodegradable polymer [30]. With a long polymer chain and positively charged groups ($-\text{NH}_3^+$), the chitosan captures and links negatively charged algal cells and other particles. Chitosan-modified clays/soils could be effective flocculants in removing HABs [31,32]. Positively charged iron on the clay could help with the attachment of algal cells, while it will react with phosphorus in water to form precipitates on the clay surface. In this study, we aimed to evaluate the efficiency of modified sepiolite and kaolinite in removing cyanobacteria and phosphate, and to determine their adsorption capacity for the algal toxin microcystin-LR (MC-LR). Furthermore, to reduce internal phosphorus cycling, the modified clays were packed into semi-permeable membrane tubes to enhance phosphate and MC-LR removal.

2. Materials and methods

2.1. Algae culture

Microcystis aeruginosa, one of the most common potentially toxic bloom-formers in eutrophic freshwater, was used as the model cyanobacteria for HAB study. *M. aeruginosa* (UTEX LB 2385) was cultured in BG-11 medium at room temperature (21 °C) with a pH of 7.8–8.2. The light source was a white fluorescent lamp with a maximum intensity of 2100 Lux, and a 12-hour dark/12-hour light periodicity was maintained. Algae were re-cultured during their exponential growth phase during the research period for culture maintenance. Algae in their exponential growth stage were used for these experiments.

2.2. Clay type selection and clay modification

Pan et al. [27] have identified the flocculation capability of 26 clays

in removing *M. aeruginosa* cells. Talc, ferric oxide, sepiolite, ferroferric oxide, and kaolinite were categorized as group I clays, based on their high 8-h equilibrium removal efficiency (93.9–97.6 %) at a clay loading of 0.7 g/L [27]. In this study, kaolinite (ACROS Organics, France) and sepiolite (Sigma-Aldrich Chemicals, Spain) were selected for the HAB control experiment after modification. Kaolinite, $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$, is a layered silicate mineral characterized by its soft consistency. Sepiolite, $\text{Mg}_4\text{Si}_6\text{O}_{15}(\text{OH})_2 \cdot 6\text{H}_2\text{O}$, is a fibrous magnesium silicate with a porous structure and a larger specific surface area.

The two clays, kaolinite and sepiolite, were modified by chitosan and iron, based on an approach in a previous study with modification [28]. First, a chitosan solution was prepared by dissolving 100 mg of chitosan in 10 mL of 1 % hydrochloric acid. Then, this solution was diluted with deionized water to reach a concentration of 1 g/L. For clay preparation, 230 mg of clay was added to 1 L of deionized water. After that, 1000 mg of $\text{NH}_4\text{Fe}(\text{SO}_4)_2$ was added to the same clay solution with 5.8 mL of previously prepared chitosan solution. Then, the solution was stirred for 5 h with a magnetic stirrer at 600 rpm. Finally, the solution was filtered, dried, and ground to obtain modified clay. The modified clay was used to remove cyanobacteria, microcystin, and phosphate.

2.3. Characterization of commercial and modified clays

Both commercial and modified clay samples were analyzed using an S-3400 N II scanning electron microscope (SEM) equipped with an Energy-Dispersive X-ray spectrometer (EDX) (Hitachi, Tokyo, Japan) to establish the clay morphologies and elemental distribution at 10.0 kV and $\times 6\text{k}$ magnification. To verify the chitosan modification on the clays, the infrared spectra for both commercial and modified kaolinite and sepiolite were obtained by the compact fourier transform infrared spectroscopy (FTIR) instrument (IRSpirit-X, Shimadzu, Tokyo, Japan) using the attenuated total reflectance (ATR) adaptation. 2–5 mg of clay was mounted under the ATR and scanned for the IR spectra in the range $500\text{--}4000\text{ cm}^{-1}$ with 50 scans per spectrum at a 8 cm^{-1} spectral resolution. Moreover, N_2 adsorption-desorption isotherms were recorded using an ASAP 2050 micropore analyzer (Micromeritics, Norcross, GA, USA) at 77 K to investigate the Brunauer, Emmett, and Teller (BET) surface area and Barrett–Joyner–Halenda (BJH) pore size distribution of both commercial and modified kaolinite and sepiolite. The average particle size and the zeta potential across the pH range 2.0–10.0 were measured in commercial and modified kaolinite and sepiolite using the Zetasizer Nano ZS instrument (Malvern Panalytical, Malvern, UK).

2.4. Experimental procedures

2.4.1. Cyanobacteria, phosphate, and microcystin removal by clay flocculation

Both commercial and modified clays, as flocculants, were tested for their capabilities of removing algal biomass, microcystin, and phosphate. The algal culture and clay mixtures were mixed at 160 rpm for 20 min, followed by a settling time of 5 min for cyanobacteria removal at room temperature. The supernatant was collected to quantify cyanobacteria inside by measuring the absorbance at 680 nm. Cultured algae solutions or diluted BG-11 media with known PO_4^{3-} concentrations were tested in duplicates for phosphate removal efficiencies of commercial clays and modified clays for 10–12 days after culturing to reveal the kinetics of phosphate removal after clay application and optimize the treatment time. The concentrations of phosphate in diluted BG-11 media ranged from 1.4 to 10.8 mg/L to simulate the waterbody near municipal wastewater discharge points, given that secondary wastewater treatment plants without an enhanced biological phosphorus removal process could result in effluent concentrations of 3–10 mg P/L (9–21 mg PO_4^{3-} /L) [33]. These experiments were conducted in duplicates.

For microcystin adsorption, a 2.5 mM (250 $\mu\text{g}/100\text{ }\mu\text{L}$) MC-LR stock solution in DMSO (Millipore Sigma, Taiwan) was purchased for preparing the MC-LR solutions. By using an MC-LR stock solution with a

concentration of 100 µg/mL, a series of MC-LR solutions with initial concentrations of 20 µg/L, 40 µg/L, 100 µg/L, 200 µg/L, and 400 µg/L were prepared for MC-LR adsorption experiments. The selection of MC-LR concentration investigated in this study was based on the commonly detected microcystin concentration ranges in natural water bodies [34–39]. A 60 mg of modified sepiolite and kaolinite was added directly per 15 mL MC-LR samples with known initial concentrations. The solutions were set on an orbital shaker at 100 rpm for 48 h to reach equilibrium. After that, the supernatant was taken to determine the MC-LR adsorption to clay using the Abnova Microcystin ELISA test kit. All the microcystin adsorption tests were conducted in triplicate.

2.4.2. Phosphate and microcystin removal by dialysis tubes with modified clays

Dialysis tubes use a semi-permeable membrane to separate molecules according to their size. These are generally made of cellulose acetate, and these tubes are primarily used in medical treatment to remove toxins from patients' bloodstreams, acting as a selective barrier. In this study, dialysis tubes with a pore size threshold of 12–14 kDa (the diameter of 1.25", Aldon Corporation, NY) were used to recover phosphate from water bodies to minimize its internal cycling. Theoretically, both MC-LR and PO_4^{3-} can freely pass through the membrane pores while clay particles are retained. Modified clay with a load of 800 mg clay/L was packed in dialysis tubes to remove phosphate from water with algae.

The modified kaolinite and sepiolite loading for the MC-LR removal experiment with clays packed inside the dialysis tubes was 4 g/L. The initial MC-LR concentrations were 100 µg/L, 200 µg/L, 400 µg/L, 600 µg/L, and 800 µg/L. The solutions were set on an orbital shaker at 100 rpm until equilibrium. Then, the supernatant was collected after 3 days, 5 days, 7 days, and 10 days to determine the MC-LR adsorption to clay. The phosphate and microcystin in water can enter the semi-permeable membrane of dialysis tubes and react with iron-modified clays. Once microcystin is adsorbed and the phosphate is precipitated on the clays, dialysis tubes can be removed from the water body.

2.5. Analytical methods

The total Iron (Fe) content in modified clays was analyzed using the Hach FerroVer Iron reagent and a benchtop spectrophotometer (DR 3900, Hach, USA). First, the adsorbed Fe^{3+} on clay was dissolved in a 0.1 M HCl solution before determining the iron concentration in the acid solution (detection range 0.02–3 mg/L Fe). The Fe content in the clays (mg Fe/g clay) can be calculated by using the total iron mass in the acid solution divided by the mass of the clays. Phosphate concentrations before and after clay treatment were measured using Hach PhosVer 3 phosphate reagent (detection limit: 0.02 mg/L) and the same spectrophotometer (DR 3900, Hach, USA) in duplicate. Absorbance at 680 nm was measured to represent cyanobacteria concentration in water, as it indicates the presence of Chlorophyll a [40]. The microcystin-LR ELISA kit (KA1496, Abnova, Taiwan) with a detection limit of 0.1 µg/L was used to test the MC-LR concentration in the liquid before and after clay application, following the manufacturer's instructions after appropriate dilutions. MC-LR standards in the kit were used to construct standard curves for measuring the unknown concentrations in the samples. For all tests, DI water was used as the blank, and chemical standards of MC-LR and phosphate were also spiked into the diluted BG-11 media in Section 2.4.1 for recovery analyses to ensure the reliability of the results. For the determination of total Iron (Fe) content and phosphate concentration, samples were collected and measured in duplicate. For the MC-LR concentration measurement, sampling and measurement were done in triplicate.

3. Results

3.1. Characterization of modified clays

In Fig. 1, SEM images of commercial kaolinite, modified kaolinite, commercial sepiolite, and modified sepiolite are presented, respectively. A distinct clay surface was observed when comparing the commercial and modified kaolinite and sepiolite SEM images. Commercial kaolinite showed a structure of stacked sheets. After the modification of kaolinite, scattered clay particles were gathered closer. Natural or commercial sepiolite has a fibrous structure, contributing to its networking and bridging effect in flocculating algae cells. The enhanced porous structure of sepiolite with the modification can be observed in Fig. 1b and d. The SEM-EDS data for modified kaolinite and sepiolite, which are provided in the supplementary data (Fig. S1), showed distinct peaks of Fe. There were no peaks observed for Fe in commercial kaolinite and sepiolite. The total Fe content in both modified and commercial clays was investigated. Commercial kaolinite had a total iron content of 0.09 mg Fe/g clay, and modified kaolinite had 1.42 mg Fe/g clay. The total iron content in commercial and modified sepiolite was 0.02 mg Fe/g clay and 1.47 mg Fe/g clay, respectively. These results indicate successful Fe incorporation into the clay. FT-IR spectrum (Fig. S2) also demonstrated the successful modification of chitosan as indicated by more prominent adsorption of N–H stretch ($3400\text{--}3250\text{ cm}^{-1}$) and the N–H bend (from 1650 to 1580 cm^{-1}).

BET surface area, Langmuir surface area, pore width, total volume in pores, total pore area, and average particle size of both commercial and modified kaolinite and sepiolite are listed in Table 1. The BET surface area of kaolinite increased with the modification from $9.60\text{ m}^2/\text{g}$ to $36.29\text{ m}^2/\text{g}$. The commercial sepiolite has a BET surface area of $143.42\text{ m}^2/\text{g}$, which increased to $167.95\text{ m}^2/\text{g}$ after the modification, whereas the Langmuir surface area remained unchanged after modification. Compared to kaolinite, the high surface area of sepiolite may be attributed to the small pores and small particle size on the surface of sepiolite (Table 1). The pore width of commercial and modified sepiolite was 84 Å , much smaller than the pore widths of commercial (232 Å) and modified kaolinite (121 Å). The average particle size for modified sepiolite was 0.8 µm , whereas the modified kaolinite was 1.23 µm . Based on the ratio of the total area in pores and Langmuir surface area, modified kaolinite and sepiolite have 63 % and 83 % of the surface area contributed by the pores, revealing a higher porosity of sepiolite compared to kaolinite. Moreover, modified sepiolite also has a significantly higher pore volume ($0.26\text{ cm}^3/\text{g}$) than kaolinite ($0.10\text{ cm}^3/\text{g}$), which may benefit the adsorption and chemical precipitation.

A study conducted by Tang et al. has published their findings that the natural attapulgite, 5.0 M and 10.0 M HCl modified attapulgite, have BET surface areas of 136.98, 89.59, and $95.94\text{ m}^2/\text{g}$, respectively [23]. The total volume in pores for natural attapulgite was $0.0061\text{ cm}^3/\text{g}$, whereas the total pore volumes for 5.0 M and 10.0 M modified attapulgite were reduced to 0.0019 and $0.0005\text{ cm}^3/\text{g}$, respectively, after the modification. Y. Tang and the team observed that 90 % of *M. aeruginosa* removal efficiency was achieved by 5.0 M HCl modified attapulgite with a clay loading of 0.37 g/L. Another research study has reported increased BET surface area from 19.4 to $47.6\text{ cm}^2/\text{g}$ and decreased pore volume from 0.104 to $0.052\text{ cm}^3/\text{g}$ after modifying kaolinite [41]. The modification process included the processes of calcination, acid etching, alkali neutralizing, and aging. It is also reported that the algal removal efficiency of 87.9 % was obtained with the modified kaolinite dose of 0.2 g/L. In this study, modification with iron and chitosan increased the surface area of both kaolinite and sepiolite. The modified sepiolite had a significantly higher total pore volume and BET surface area than the materials used in other studies.

The zeta potential data for commercial and modified kaolinite and sepiolite are plotted in Fig. 2. The zeta potential value for commercial kaolinite was negative in the pH range 2–10. Kaya et al. have reported the same observation for the zeta potential values for kaolinite [42].

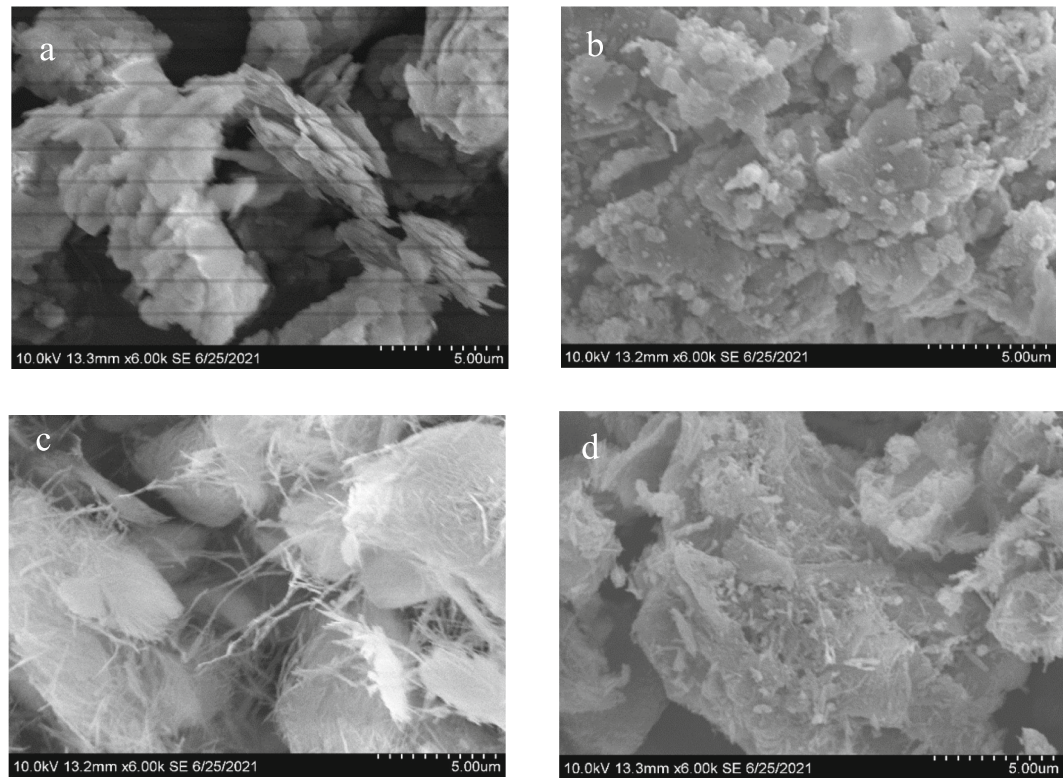


Fig. 1. SEM Images of a-Commercial Kaolinite, b-Modified Kaolinite, c-Commercial Sepiolite, d-Modified Sepiolite.

Table 1
Characteristics of commercial and modified kaolinite and sepiolite.

Properties	Commercial kaolinite	Modified kaolinite	Commercial sepiolite	Modified sepiolite
BET Surface Area (m ² /g)	9.60	36.29	143.42	167.95
Langmuir Surface Area (m ² /g)	10.91	32.46	112.72	110.29
Pore width (Å)	232.06	121.79	84.59	84.21
Total Volume in Pores (cm ³ /g)	0.01	0.10	0.27	0.26
Total Area in Pores (m ² /g)	6.20	20.60	110.60	91.68
Average particle size (μm)	1.40	1.23	0.96	0.80
Particle size range (μm)	1.28–1.99	0.96–1.48	0.62–1.28	0.53–0.96

Modified kaolinite exhibited an increased zeta potential, but it still showed negative zeta potential values, except in the pH range of 5.0 to 6.5, where the zeta potential became positive. For commercial sepiolite, the zeta potential was +3.96 mV at pH 2.0. The isoelectric point (IEP) or the pH value with zero zeta potential was around pH = 2.5, and beyond that, up to pH = 10.0, the zeta potential was negative. Modified sepiolite had a more positive zeta potential compared with commercial sepiolite at all pHs and showed a positive zeta potential up to pH = 7.4. The zeta potential values for modified sepiolite in our study were comparable to Fe-modified sepiolite in the previous study [28]. The more positively charged clays could benefit the flocculation of cyanobacteria cells through electrostatic neutralization [31].

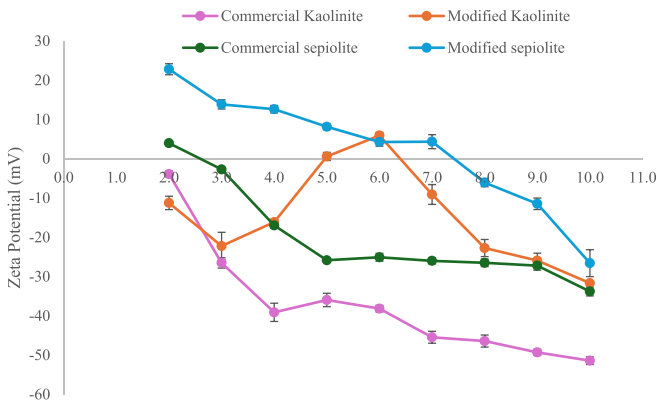


Fig. 2. The zeta potential of commercial and modified clays ranging from pH 2 to pH 10.

3.2. Removal of Cyanobacteria by clay flocculation

Kaolinite and sepiolite were tested for their ability to remove cyanobacteria. Both modified and commercial clays of sepiolite and kaolinite were added to lab-cultured *M. aeruginosa* samples and flocculated for 20 min. After settling for 5 min, the samples' supernatants were collected to measure absorbance at 680 nm as an indicator of algal cell number. Fig. 3 shows the absorbance variation at 680 nm for the control, 2 g/L, and 5 g/L clay loading for modified and commercial clays. The samples with commercial clay inside were cloudy, which explains the increase in absorbance at 680 nm. The clay modification promoted the aggregation and precipitation of cyanobacterial cells. It was observed that the absorbance at 680 nm for both modified clays was reduced when flocculated with clay. The absorbance at 680 nm further dropped when clay loading increased from 2 g/L to 5 g/L for modified clays, suggesting a significant increase in algal cell removal efficiency with

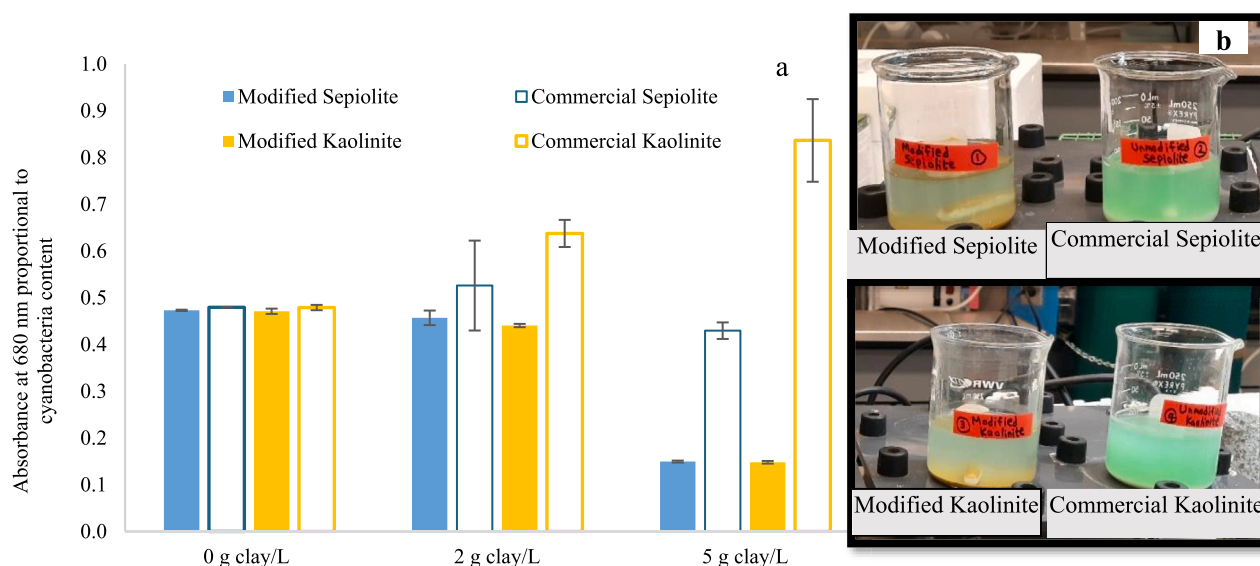


Fig. 3. The variation of cyanobacteria content with modified and commercial clay loading, a: Graphical representation, b: Flocculation with clay (2 g/L and 5 g/L).

increased clay loading. There was no significant difference between modified sepiolite and kaolinite in removing *M. aeruginosa* cells (p -value = 0.154). All the algae were settled at the bottom of the beakers with the modified sepiolite and kaolinite. About 68 % of cyanobacteria removal efficiency was obtained based on the absorbance reduction after 20 min of clay flocculation with 5 g/L clay loading for modified kaolinite and sepiolite. After adding modified clays, the pH of the algal solution dropped to 6.4–7.3. Under this pH range, both modified kaolinite and sepiolite had a positive zeta potential, which enhanced the flocculation of *M. aeruginosa* through electrostatic neutralization [31]. That explains the comparable performance of both modified kaolinite and sepiolite in flocculating algal cells. In contrast, the commercial kaolinite and sepiolite were more negatively charged than -25 mV, and the clays themselves could not destabilize and aggregate, causing the increased turbidity [31]. *M. aeruginosa* cells float on water surfaces because of the gas vesicles available in their cell structure. Clay and algae are negatively charged naturally [15,28]. The surface modification of the clay by Fe^{3+} ions made it positively charged and caught negatively charged algal cells more effectively. Attached algal cells settled to the bottom of the water together with clay, removing algal cells from the water's surface.

Chitosan is a biopolymer with long polymer chains, netting, and bridging features of aggregates related to coagulants and flocculants [28,43]. Chitosan also has an immense amount of positively charged amine groups in its structure. It promotes the electrostatic interactions between chitosan polymer chains and negatively charged algal cells, expediting the algal cell trapping and removal together with clay [44]. More algal cells were captured and trapped in this network created by chitosan. Clay works as a ballast to this network linked by chitosan and settles the scavenged algal cells [28]. The Fe^{3+} and chitosan coating enhanced the coagulant properties of clays by adding positive charges and bridging and netting features.

According to a study conducted by Pan et al. (2006), *M. aeruginosa* removal efficiency of more than 90 % was obtained after 8 h of flocculation with kaolinite and sepiolite at a clay loading of 0.7 g/L [27]. At the same time, sepiolite has shown a consistent removal efficiency of around 90 % with varying clay loading of 0.2 g/L to 0.7 g/L in contrast to kaolinite [27]. However, flocculation experiments conducted by Pan et al. had much lower initial *M. aeruginosa* concentrations (optical density (OD) at 680 nm = 0.1 or 4.86×10^6 cells/mL). In this study, the high initial cyanobacteria concentration ($\text{OD}_{680} = 0.48$) resulted in the incompetence of commercial kaolinite and sepiolite. Pan's study also

indicated that flocculation equilibrium was not reached until 240 min. Type I clays, including kaolinite and sepiolite, removed only 60 % of cyanobacteria after 20 min. Extended flocculation times are expected to improve the removal efficiency further. HAB typically has cyanobacteria of 20,000 to 100,000 cells/mL [45], which is 4 log lower than the initial cyanobacteria number used in this study. Therefore, high algal removal efficiency is also anticipated when the modified sepiolite is used to control the HAB in the real world.

3.3. Removal of phosphate by clay flocculation

Initially, the behavior of phosphate removal with different clay loadings was studied with modified and commercial clays. With an initial phosphate concentration of 11 mg/L, phosphate concentrations after 20 min with clay flocculation were measured (Fig. 4a). The phosphate removal efficiencies of commercial sepiolite and kaolinite were negligible at every clay loading. The maximum phosphate removal efficiency observed for commercial sepiolite and commercial kaolinite were 3.3 % and 2.9 %, respectively. An increasing correlation between clay dose and phosphate removal efficiency highlights that phosphate removal was higher when the applied clay loading was higher. The phosphate removal efficiency of sepiolite was higher than that of kaolinite at all clay loadings. The difference between modified kaolinite's and modified sepiolite's removal efficiencies was reduced with increased clay loading. The maximum phosphate removal efficiencies for modified sepiolite and modified kaolinite, at 96 ± 0.4 % and 96 ± 0.7 %, respectively, were achieved with a 2.0 g/L clay loading. The PO_4^{3-} removal was mainly because of the chemical reaction between PO_4^{3-} in the bulk solution and the Fe^{3+} ion attached to the modified clay ($\text{Fe}^{3+} + \text{PO}_4^{3-} + 2\text{H}_2\text{O} \rightarrow \text{FePO}_4 \cdot 2\text{H}_2\text{O} \downarrow$) [46], although adsorption also plays a significant role [13]. Based on the total iron content in modified sepiolite (1.47 mg Fe/g clay), the maximum phosphate removal should be 5 mg/L with 2.0 g/L clay loading. Given that 10.5 mg/L of phosphate was removed (96 % removal), adsorption by modified sepiolite also contributed to 50 % removal. When the clay loading was higher, more iron could be used for the precipitation of iron phosphate, thereby increasing phosphate removal efficiencies. When the clay dose was low, the enhanced phosphate removal by modified sepiolite may be attributed to the higher surface area compared with modified kaolinite, which promotes the contact of phosphate with iron immobilized on the clays. There was no significant difference between the PO_4^{3-} removal efficiencies of modified kaolinite and sepiolite (p -value = 0.67, ANOVA

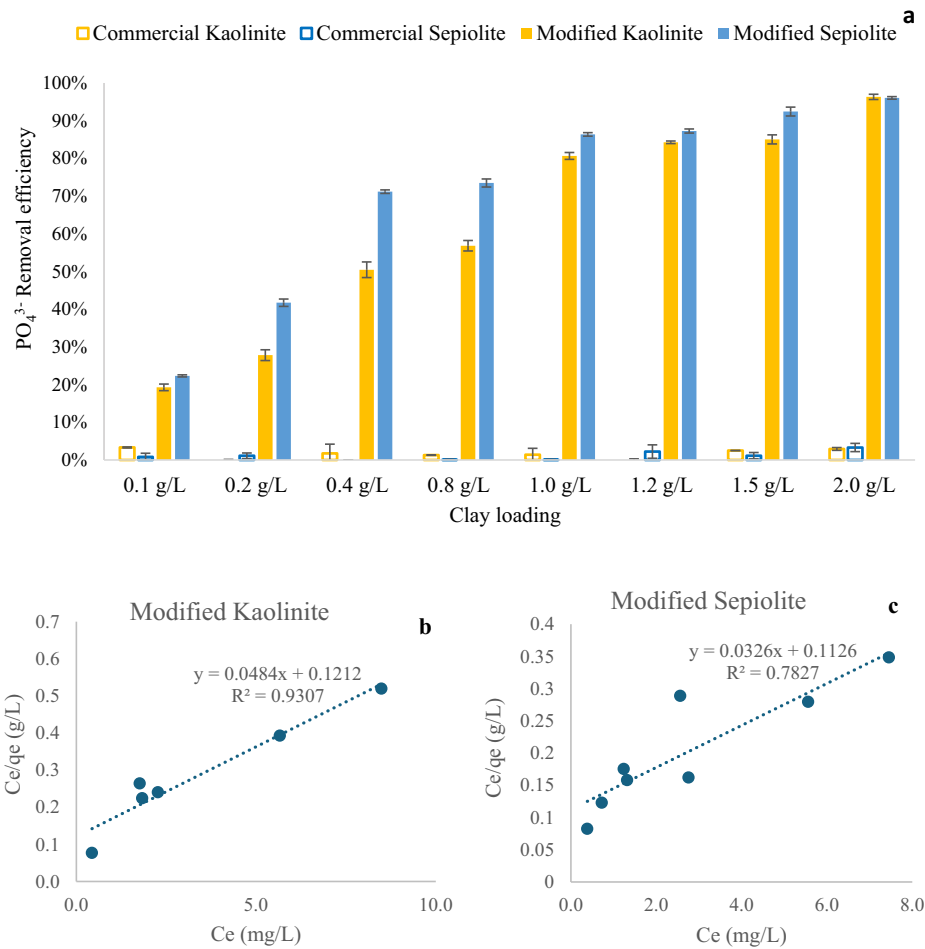


Fig. 4. a. Variation of phosphate removal efficiency by commercial and modified kaolinite and sepiolite with different loads; b. Langmuir adsorption isotherm for modified kaolinite. c. Langmuir adsorption isotherm for modified sepiolite.

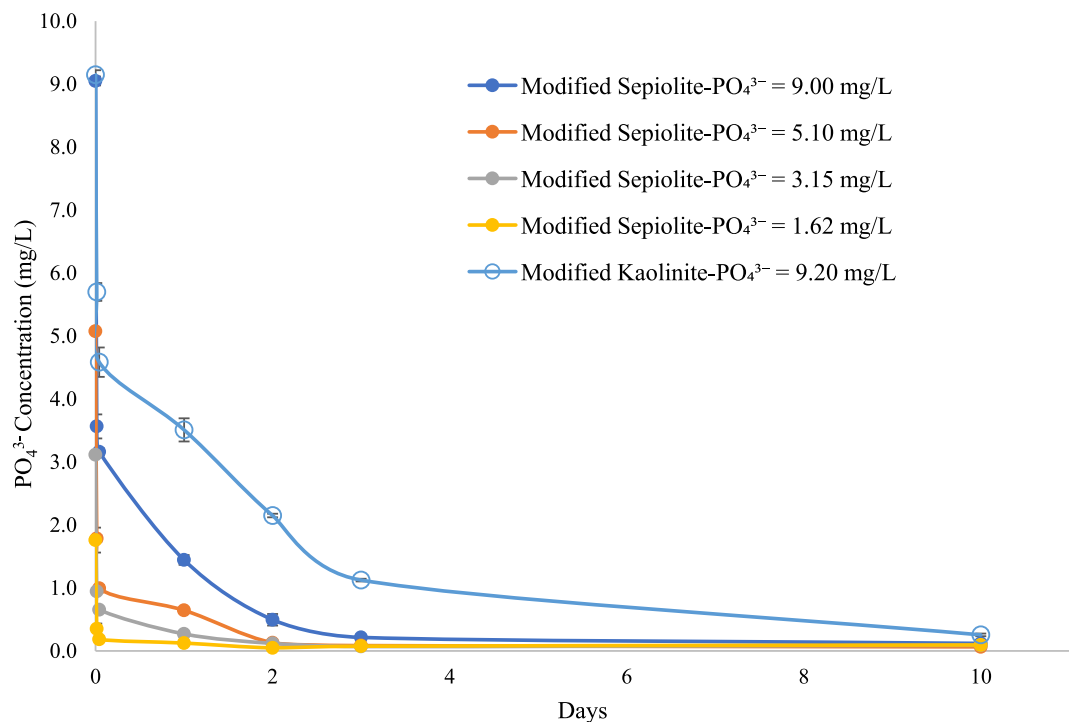


Fig. 5. The variation of PO₄³⁻ concentration over time with different initial PO₄³⁻ concentrations for modified clays at a clay loading of 0.8 g/L.

single factor) when clay loading was 2.0 g/L. PO_4^{3-} removal by modified clay fits the Langmuir isotherm well (Fig. 4b and c). The maximum capacities of modified kaolinite and sepiolite for PO_4^{3-} adsorption were 20.7 and 30.7 mg PO_4^{3-} /g clay.

With a clay loading of 0.8 g/L, PO_4^{3-} concentrations were also measured in 20 min, 1 h, 24 h, 48 h, 72 h, and 240 h after modified clay application to PO_4^{3-} solutions to study the kinetics of phosphate removal (Fig. 5). It showed that most PO_4^{3-} was removed within 20 min with direct modified clay flocculation. After 20 min, 58.9 % and 37.0 % of PO_4^{3-} were removed with modified sepiolite and kaolinite, respectively. After 24 h of direct modified clay flocculation, 84.6 % of PO_4^{3-} was removed with modified sepiolite, while 60.4 % was removed with modified kaolinite. This observation also implies that modified sepiolite excels in PO_4^{3-} removal compared to modified kaolinite in direct clay flocculation. Afterward, the removal of phosphate slowed down, and the same trend of PO_4^{3-} removal with time could be observed for all initial concentration conditions. When modified kaolinite and modified sepiolite were compared, modified sepiolite indicated a higher PO_4^{3-} removal than modified kaolinite within the contact time when initial concentrations were similar (Fig. 5). The equilibrium was reached after 72 h for modified sepiolite, whereas it took 240 h for modified kaolinite to reach the equilibrium, indicating the more rapid phosphate adsorption by modified sepiolite.

Modified clays can strip dissolved P from the water column and hold the P release from the sediment after settling [47,48]. It was reported that local red soil with a composition dominated by kaolinite and some iron and aluminum oxides had a maximum phosphate adsorption of 3.0 mg/g [13]. This local red soil had 10.8 % of Fe_2O_3 in its composition. The higher adsorption is likely due to the greater goethite content in these soils [13]. Singh and Gilkes (1992) estimated the maximum P sorption values for kaolinite and goethite ($\text{FeO}(\text{OH})$) as 0.5 mg/g and 8 mg/g, respectively, suggesting Fe content in adsorbents might have contributed mainly to the phosphate removal in water [49]. In a different research, lanthanum-modified bentonite Phoslock had maximum phosphate adsorption of 12.3 mg/g clay under oxic conditions [13]. In this study, the maximum phosphate adsorption of 30.7 mg/g modified sepiolite was superior to the literature data mentioned above. The high adsorption capacity of modified sepiolite in our study implies the success of the proposed modification to clays and the potential of using this modified clay to remove phosphate from water bodies, considering the low cost of iron modification compared with lanthanum modification.

3.4. Removal of microcystin by clay flocculation

Microcystin-LR, is considered the most widespread and toxic of the identified microcystins. The microcystin MC-LR was selected for the adsorption experiments to investigate the effectiveness of commercial and modified clays in removing algal toxins. Commercial kaolinite or sepiolite showed no MC-LR removal during the clay flocculation. It was observed that MC-LR removal efficiency increased with the initial MC-LR concentration (Fig. 6) for modified clays. MC-LR removal efficiency by modified sepiolite rose from 95.5 % to 99.9 % when the initial concentration increased from 20 $\mu\text{g/L}$ to 200 $\mu\text{g/L}$. There was a sudden drop in MC-LR removal efficiency to 96.7 % when the initial MC-LR concentration increased to 400 $\mu\text{g/L}$ (Fig. 6), with the adsorption capacity of 96.72 $\mu\text{g MC-LR/g}$ modified sepiolite. Although high concentrations of MC-LR can enhance the diffusion of ions onto the clay surface, the adsorption capacity may stabilize as the active sites of the adsorbent become saturated, resulting in a reduced removal rate [50,51]. When modified kaolinite with initial concentrations of 20 $\mu\text{g/L}$ and 40 $\mu\text{g/L}$ was used for MC-LR adsorption, no MC-LR adsorption to clay was detected. The highest MC-LR adsorption of 87.9 % was achieved for the MC-LR solution with an initial concentration of 400 $\mu\text{g/L}$, while the adsorption was 71 % for an initial MC-LR concentration of 100 and 200 $\mu\text{g/L}$.

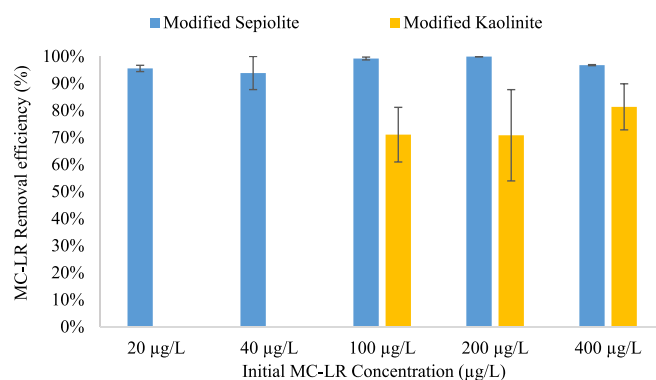


Fig. 6. The variation of MC-LR Removal by direct clay flocculation of modified sepiolite and kaolinite for different initial MC-LR concentrations,

It was reported that the total microcystin concentrations in Western Lake Erie were 0.068–14.88 $\mu\text{g/L}$ in 2016 and 0.050–10.15 $\mu\text{g/L}$ in 2017, although microcystin concentrations in Maumee River running into Lake Erie ranged from 0.17 to 305.03 $\mu\text{g/L}$ [34,35]. The total microcystin concentration was 11.8 $\mu\text{g/L}$ at a water treatment plant inlet in Western Lake Erie in 2019 due to a CHAB developed in southwestern Lake Erie [36]. MC-LR concentrations of several drinking water reservoirs reported in the literature are 2.47 $\mu\text{g/L}$ in Quebec Water Reservoir in Canada [37], 0.92 $\mu\text{g/L}$ in James River Estuary in Virginia, USA, [38], and 3.0 $\mu\text{g/L}$ in the Cascata water reservoir in Brazil [39]. Modified kaolinite cannot efficiently remove microcystin, given that the concentrations are typically low in most locations. The World Health Organization (WHO) has recommended the drinking water guideline value of 1 $\mu\text{g/L}$ for MC-LR for lifelong ingestion [52]. Modified sepiolite could reduce the concentration of MC-LR to 0.02–0.90 $\mu\text{g/L}$ when the initial concentration ranged from 20 to 400 $\mu\text{g/L}$.

In a previous study, chitosan-modified local soil was reported to reduce MC-LR concentration by 31 % (from 15.45 $\mu\text{g/L}$ to 10.64 $\mu\text{g/L}$) at a soil load of 100 mg/L, which brought about the adsorption capacity of 48.1 $\mu\text{g MC-LR/g}$ modified soil [43]. Bentonite modified with the bromide salt of the organic cation ODTMA (octadecyltrimethyl ammonium) had shown approximately 100 % removal of MC-LR at 1 g/L of modified bentonite, and that can be interpreted as 100 $\mu\text{g MC-LR/g}$ modified bentonite when initial MC-LR concentrations were lower than 100 $\mu\text{g/L}$ and lower removal rates (< 60 %) at higher initial concentrations (200 and 400 $\mu\text{g/L}$) [53]. In our study, the adsorption capacity of MC-LR was 96.72 $\mu\text{g MC-LR/g}$ modified sepiolite, comparable with the reported value but with a higher clay dose of 4 g/L. Despite lower removal rates (< 60 %) being observed at higher initial concentrations (200 and 400 $\mu\text{g/L}$) in the same study [47], it was not observed in our study with modified sepiolite. About 96.7 % of MC-LR removal was observed for an initial MC-LR concentration of 400 $\mu\text{g/L}$ with modified sepiolite at a dose of 4 g/L. The modified sepiolite used in this study has shown greater performance with a wide range of initial microcystin concentrations, indicating its potential in field applications.

3.5. Removal of phosphate and microcystins by clay inside semi-permeable tubes

Dialysis tubes use a semi-permeable membrane to separate molecules according to their size. In this study, dialysis tubes with packed clays inside are used to remove both phosphate and microcystin water, thereby minimizing internal cycling of phosphorus. Phosphate and microcystin can enter the semi-permeable membrane of dialysis tubes by diffusion and react with iron-modified clays packed inside. The PO_4^{3-} removal performance of dialysis tubes packed with commercial and modified sepiolite was compared. With a constant clay loading of 800 mg/L, the change of PO_4^{3-} concentrations with time is shown in Fig. 7. It

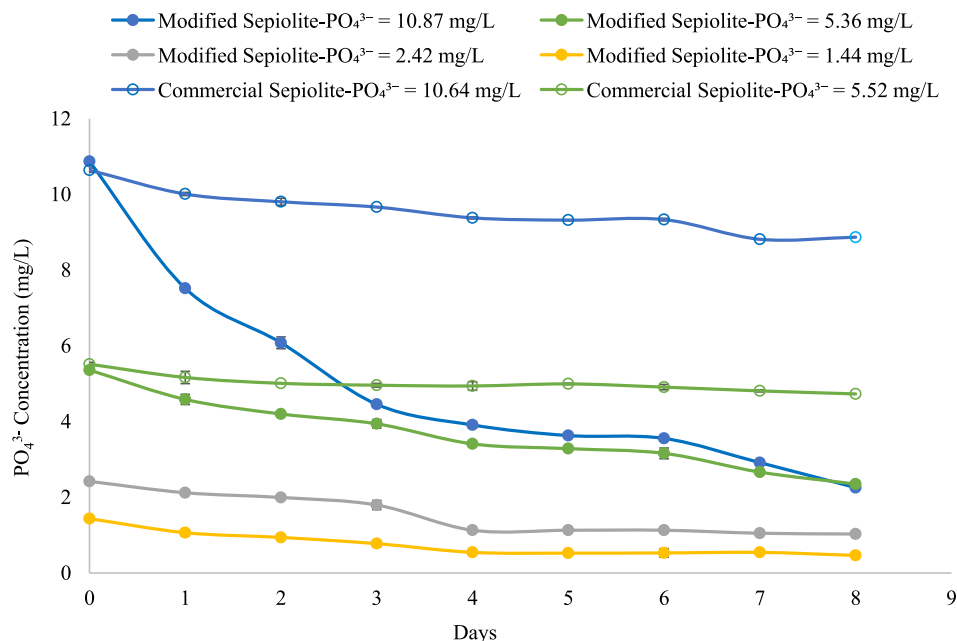


Fig. 7. Variation of PO_4^{3-} concentration over time for different initial PO_4^{3-} concentrations with modified and commercial sepiolite inside the semi-permeable membrane.

was observed that PO_4^{3-} removal was higher when the initial PO_4^{3-} concentration was higher, regardless of the clay type. Compared with direct clay flocculation, the PO_4^{3-} removal by the clays packed in the dialysis tubes was much slower. The speed-limiting step was the PO_4^{3-} diffusion across the semi-permeable membrane of dialysis tubes. The modified clays in the dialysis tubes became aggregated and settled quickly after being mixed and inserted into the tubes. The aggregated clay may impact the diffusion of phosphate and MC-LR. The high PO_4^{3-} concentration could accelerate the diffusion and lead to high removal. The highest PO_4^{3-} removal efficiency of dialysis tubes with modified sepiolite was 79 % after 8 days of treatment at the initial PO_4^{3-} concentration of 10.87 mg/L. In contrast, the highest PO_4^{3-} removal efficiency for dialysis tubes packed with commercial sepiolite was only 16.6 % at an initial PO_4^{3-} concentration of 10.64 mg/L after 8 days of treatment. A maximum PO_4^{3-} removal efficiency of 99 % was achieved after 14 days of flocculation of modified sepiolite when the initial PO_4^{3-} concentration was 10.87 mg/L.

The MC-LR adsorption experiments were conducted using modified sepiolite and modified kaolinite. Commercial clay was not used for dialysis tube arrangement as it did not show any removal of MC-LR, even with direct clay flocculation. When the initial MC-LR concentrations were 20 $\mu\text{g/L}$ and 40 $\mu\text{g/L}$, the modified clays in the dialysis tube could not remove MC-LR. For higher initial MC-LR concentrations, the removal of MC-LR was not observed until day 3 with modified clays. The MC-LR removal by modified kaolinite packed inside the dialysis tubes was negligible. Fig. 8 shows the variation of MC-LR concentration over time for different initial MC-LR concentrations with modified sepiolite. Similar to the PO_4^{3-} removal, high MC-LR concentration in the bulk solution could facilitate diffusion across semi-permeable dialysis tubes, leading to high MC-LR removal. The highest MC-LR removal efficiency of 97 % was achieved at the highest initial MC-LR concentration of 800 $\mu\text{g/L}$ applied. An average of 90 % MC-LR removal efficiency could be achieved by modified sepiolite inside the dialysis tube after 10 days.

When the clay-based method used for HAB remediation is applied, it is often applied directly to the affected water bodies. However, adsorbed P is released back into the water column, rekindling the bloom with time. Therefore, this innovative solution of using semi-permeable membrane tubes with modified clay inside was developed for phosphate recovery from water bodies to minimize the internal cycling of

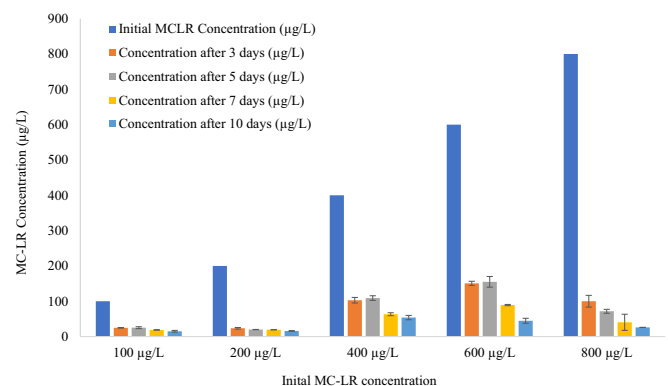


Fig. 8. Variation of MC-LR concentration over time for different initial MC-LR Concentrations with modified sepiolite inside semi-permeable membrane.

phosphate. Once microcystin is adsorbed and the phosphate is precipitated on the clays, dialysis tubes can be removed from the water body. Semi-permeable membrane modules with modified clays inside are proposed to submerge in the water with an external floating arrangement (Fig. S3). It is also crucial to note that since dialysis tubes are biodegradable, non-toxic, and harmless to the environment and organisms, they can be used in aquatic systems where algal blooms occur. Given that this approach is only effective when phosphate and MC-LR concentrations are high, these dialysis tubes with clays are only recommended for emergent use in lakes with severe HAB. More studies are needed to optimize the materials and pore size of semi-permeable membranes to increase the diffusion rate of phosphate and MC-LR and address the concern of the mechanical strengths. The mechanisms for removing phosphate and MC-LR require further exploration to understand the role of clay modification. Furthermore, a pilot-scale study using water columns is needed to investigate the feasibility in real-world applications.

4. Conclusion

This study modified kaolinite and sepiolite with chitosan and iron

(III) to control HABs. Compared with modified kaolinite, modified sepiolite demonstrated superiority in microcystin and phosphate removal with 99 % and 97 % removal rates, respectively. When the semi-permeable membrane tubes with modified clays were employed, phosphate and microcystin concentrations were also reduced gradually with time, indicating their potential to prevent the recurrence of HABs and mitigate the impact of existing HABs. Our results demonstrated that modified clays with semi-permeable membrane tubes could be a potential solution for mitigating harmful algal blooms by adsorbing algal toxins and removing phosphorus in water to prevent future blooms. Since this method could be used for phosphorus removal and algal toxin reduction without potential environmental risk, it becomes a more appealing solution for mitigating and preventing algal blooms.

CRedit authorship contribution statement

W.K.N.L. Abeykoon: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Yanyan Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

Yanyan Zhang reports financial support was provided by US Bureau of Reclamation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was financially supported in part by the U.S. Bureau of Reclamation [Agreement No. R20AC00023] and New Mexico Water Resources Research Institute student grant.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.algal.2025.104302>.

Data availability

Data will be made available on request.

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